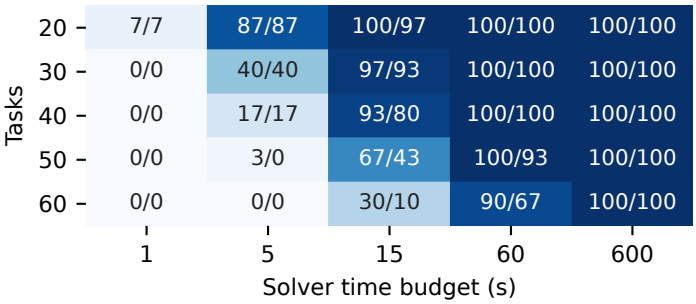
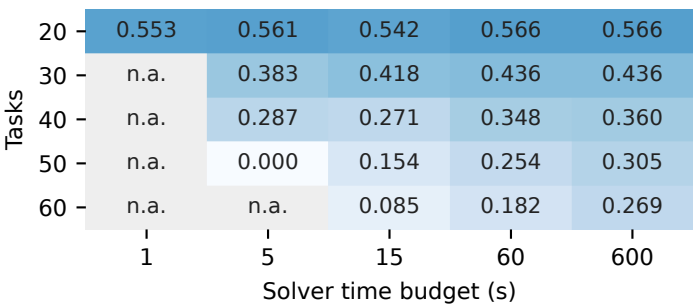
Availability, incumbent quality, and optimality certification

Experiment A (4 × 4)

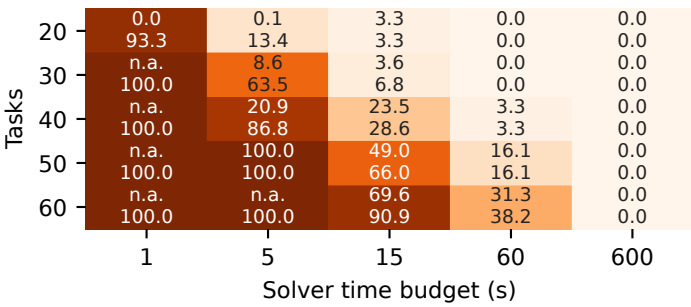
Feasible incumbent / positive plan (%)



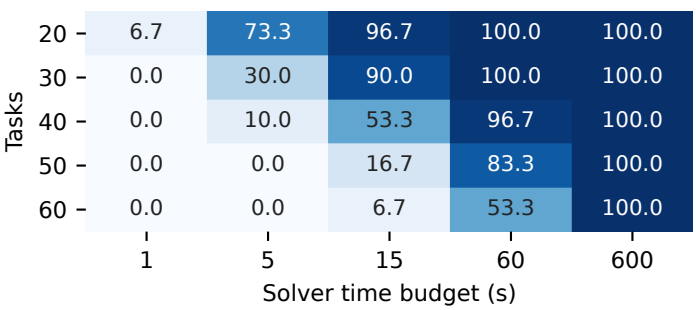
Mean best incumbent (recorded solutions)



Mean shortfall (%) — incumbent / available plan

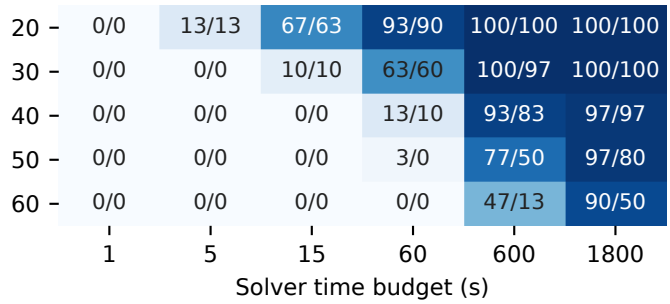


Certified optimality (%)

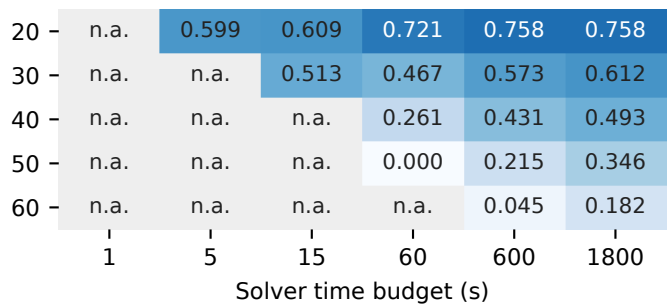


Experiment B (6 × 7)

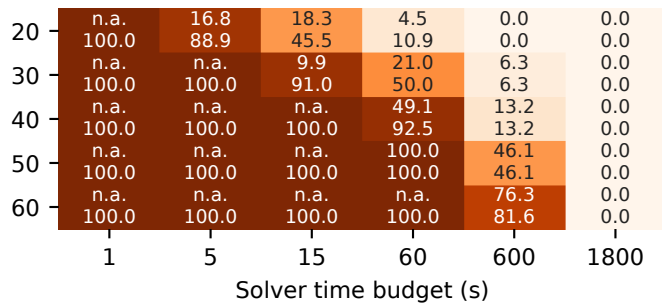
Feasible incumbent / positive plan (%)



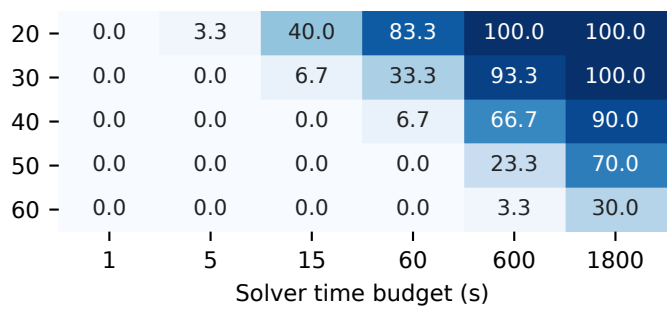
Mean best incumbent (recorded solutions)



Mean shortfall (%) — incumbent / available plan



Certified optimality (%)



30 instances per task count. First row: feasible incumbent / positive plan.
Mean incumbent excludes missing solutions (n is reported in the CSV). Shortfalls (upper/lower): incumbent / available plan, with idle=0 when no incumbent exists. Colour shows available-plan shortfall.
Certified-reference samples: A, n=30 at every size;
B, n=30, 30, 27, 21, 9 for 20–60 tasks. Budgets are independent restarts, not one continuous run.